## Multiple Choice:

1. Which of the following lines is parallel to the line $3x – 2y + 6 = 0$?

   A. $3x + 2y – 12 = 0$
   B. $4x – 9y = 6$
   C. $12x + 18y = 15$
   D. $15x – 10y – 9 = 0$

2. What is the equation of the line joining the points (3,-2) and (-7, 6)?

   A. $2x + 3y = 0$
   B. $4x – 5y = 22$
   C. $4x + 5y = 2$
   D. $5x + 4y = 7$

3. Find the distance between the given lines $4x – 3y = 12$ and $4x – 3y = -8$.

   A. 3
   B. 4
   C. 5
   D. 6

4. Given the equation of a parabola $3x + 2y^2 – 4y + 7 = 0$. Locate its vertex.

   A. (5/3, 1)
   B. (5/3, -1)
   C. (-5/3, -1)
   D. (-5/3, 1)

5. Compute the focal length and the length of the latus rectum of the parabola $y^2 + 8x – 6y + 25 = 0$.

   A. 2, 8
   B. 4, 16
   C. 16, 64
   D. 1, 4

6. The lengths of the major and minor axes of an ellipse are 10 m and 8 m, respectively. Find the distance between the foci.

   A. 3
   B. 4
   C. 5
   D. 6

7. An earth satellite has an apogee of 40,000 km and a perigee of 6,600 km. Assuming that the radius of the earth as 6,400 km, what will be the eccentricity of the elliptical path described by the satellite with the center of the earth at one of the foci?

   A. 0.46
   B. 0.49
   C. 0.52
   D. 0.56

8. Find the eccentricity of the curve $9x^2 – 4y^2 – 36x + 8y = 4$.

   A. 1.80
   B. 1.92
   C. 1.86
   D. 1.76

9. The segment from (-1,4) to (2,-2) is extended three times its own length. The terminal point is ______.

   A. (11,-24)
   B. (-11,-20)
   C. (11,-18)
   D. (11,-20)

10. Find the angle formed by the lines $2x + y –8 = 0$ and $x + 3y + 4 = 0$.

    A. 30°
    B. 35°
    C. 45°
    D. 60°

11. Find the coordinates of a point equidistant from (1, -6), (5, -6) and (6, -1).

    A. (2, -2)
    B. (3, -2)
    C. (3, -3)
    D. (2, -3)

12. What is the equation of the asymptote of the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$?

    A. $2x – 3y = 0$
    B. $3x – 2y = 0$
    C. $2x – y = 0$
    D. $2x + y = 0$

13. What is the length of the latus rectum of $4x^2 + 9y^2 + 8x – 32 = 0$?

    A. 2.5
    B. 2.7
    C. 2.3
    D. 2.9

14. Given a parabola $( y – 2 )^2 = - 8(x – 1)$. What is the equation of its directrix?

    A. x = -3
    B. x = 3
    C. y = -3
    D. y = 3

15. A parabola has its focus at (7,-4) and directrix y = 2. Find its equation.

    A. $x^2 + 12y – 14x + 61 = 0$
    B. $x^2 – 14y + 12x + 61 = 0$
    C. $x^2 – 12x + 14y + 61 = 0$
    D. None of the above

16. What is the shortest distance from A (3,8) to the circle $x^2 + y^2 + 4x – 6y = 12$?

    A. 2.1
    B. 2.3
    C. 2.5
    D. 2.7

17. Find the polar equation of the circle of radius 3 units and center at (3, 0).

    A. $r = 3 \cos \theta$
    B. $r = 3 \sin \theta$
    C. $r = 6 \cos \theta$
    D. $r = 9 \sin \theta$

18. What is the area enclosed by the curve $9x^2 + 25y^2 – 225 = 0$?

    A. 47.1
    B. 50.2
    C. 63.8
    D. 72.3

19. A line passes thru (1,-3) and (-4, 2).What is the equation of the line in slope-intercept form?

    A. $y - 4 = x$
    B. $y = -x - 2$
    C. $y = x - 4$
    D. $y - 2 = x$

20. Determine the coordinates of the point which is three-fifths of the way from the point (2,-5) to the point (-3,5).

    A. (-1,1)
    B. (-2,-1)
    C. (-1,-2)
    D. (1,-1)

21. Find the area of the triangle which the line $2x - 3y + 6 = 0$ forms with the coordinate axis.

    A. 3
    B. 4
    C. 5
    D. 2

22. If the discriminant of a quadratic equation is greater than zero, the graph is a/an ______.

    A. circle
    B. parabola
    C. ellipse
    D. hyperbola

23. What refers to a locus of a point which move so that it is always equidistant from a fixed point (focus) and from a fixed straight line (directrix)?

    A. Circle
    B. Ellipse
    C. Parabola
    D. Hyperbola

24. Find the equation of a straight line with a slope of 3 and a y-intercept of 1.

    A. $3x + y - 1 = 0$
    B. $3x - y + 1 = 0$
    C. $x + 3y + 1 = 0$
    D. $x - 3y - 1 = 0$

25. How far from the y-axis is the center of the curve $2x^2 + 2y^2 + 10x - 6y - 55 = 0$?

    A. 2.5 units
    B. 3.0 units
    C. 2.75 units
    D. 3.25 units

END